

nance isn't really an issue.

Which open source tools do you use?

We use GCC native and cross compilers, graphical libraries like GTK+ and QT for UI designs, Linux kernel for X86, ARM and Power PC in our embedded designs. We use MySQL and Postgres for database management and Fedora/Ubuntu as our desktop operating systems.

What products has AMI developed using open source?

Some of our major products use open source components. For example, our remote server management cards are based on Embedded Linux and so are some of our storage products. We have also ported versions of Linux to custom architectures and are involved in GCC compiler development. Further, one of our recent products, called power manager, is based on open source. We also have our own testing tool for embedded applications for which we have consumed open source.

In a nutshell, about 70 per cent of our applications and products developed are based on open source.

How do your developers contribute back to the community?

Our developers have been involved in kernel development and fixes in ACPI and x86 HPET, Intel i915 Display Driver/Linux Direct Rendering Manager (DRM) and Kernel PCI and SMP code bases.

What challenges do you face in using open source for project development?

The main challenge for us is version mismatches. Very often, we develop an application using the current version of the open source software, but by the time the development is completed the versions get upgraded. So we have to go back to the old setup and make the changes. This process is time-consuming, but with ample help available from the online community, we usually overcome the issue.

Talent is another challenge. The applications that we develop are in the embedded area and thus we need professionals having knowledge about Linux, 'C', hardware and microcontrollers. We impart internal training to overcome this hiccup.

Can you please elaborate on the training imparted to professionals? Do you offer upgradation programmes?

We give a complete training of three months to fine-tune the developers' skillset in microcontrollers, hardware, GUI and open source programming languages. Further, we en-

sure that the learning process is on thereafter. For instance, whenever a new distribution is released, we download it and allow the developers to experiment with it. We strongly believe hands-on training in open source technologies is the best way to upgrade one's skillset.

Do you have a dedicated team for development of embedded solutions using open source technology?

AMI works on next-generation technologies and has multiple teams (storage group, utilities group for hardware related diagnostics, embedded group, mobile app group to help in porting of Android/open source OS for netbooks and cellphones, and remote management group) involved in such development. Most of the technologies that we are developing will be available in the market six-eight months from now and most of them are built using open source. All our teams are well equipped with open source skillset. In India alone, we have about 200 people developing applications using open source.

If you build an application using Linux, no associated costs are involved when you ship the embedded hardware.

But, if the application is built on proprietary software, apart from one-time investment in the software, the upgrade cost needs to be paid on mobile devices too

What are your hiring plans for 2010-11? What kinds of skills are you looking for?

We had hired 60-70 people last year and this year too, we plan to hire another 70-80 people, well-equipped with open source and embedded system skillset. We are looking for professionals having a good engineering back-

ground, either in electronics or computer science, and good understanding of microcontrollers, 'C' and Assembly.

In the future, which projects would you pursue?

We are going big on Android and mobile application development. Thus we are looking for more projects in the mobile application space. We are also looking for projects involving embedded systems and Linux, as embedded will remain our prime focus.

What advice do you give to IT companies and developers looking to enter the open source world?

Actually, for any player in the embedded market today, Linux is the first choice. The only thing one needs to learn is how to use one's resources effectively.

Developers looking at a career in embedded systems using open source should focus on fundamentals. It is important to understand the hardware architecture and the operating system that you want to port to a particular hardware. 'One size fits all' methodology doesn't work in the embedded space. So don't design one software and force it upon the customers. Understand what solution is required and then decide on the hardware, OS and applications. ●